

CSE Alpha Engine

An automated equity research, valuation and decision-support platform
for the Colombo Stock Exchange

Document	Consolidated master specification. Supersedes <i>Architecture v1.0</i> and <i>Part II — Decision Layer v1.0</i> , which are merged in full and extended.
Companion	<i>CSE Alpha Engine — User Interface & Experience Specification v1.0</i> (separate document)
Scope	All ~286 CSE-listed equities. Every name valued, scored and explained; a gated subset marked investable.
Horizon	12–36 months. Long-term investing. Macro regime and company outlook dominate.
Design principle	Arithmetic is deterministic code. AI interprets, narrates and proposes — it never computes a valuation and never places an order.
Date	9 August 2026
Status	Build-ready. Open parameters listed in Part O.

Reference frameworks: Damodaran (NYU Stern) valuation architecture · CFA Institute Research Foundation, *Equity Valuation: Science, Art, or Craft?* (2017) · Fama–French / Carhart factor methodology · Piotroski, Beneish, Sloan, Altman, Novy-Marx accounting signals · Pesaran–Shin–Smith ARDL bounds testing · Bailey & López de Prado validation protocol.

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0 • Executive Summary

The CSE Alpha Engine is a continuously-running web application that reads every listed Sri Lankan company, values it through several independent models, prices it against the macro environment, and tells you the exact price below which each one becomes worth owning — showing its working at every step.

The analogy

Think of it as a **weather station attached to a property surveyor attached to an auctioneer**. The weather station (macro engine) tells you what season the market is in and therefore how much you should own at all. The surveyor (valuation engine) tells you what each building is actually worth, using the method appropriate to that kind of building — you don't value a bank the way you value a tea estate. The auctioneer (price engine) tells you the highest bid you can make and still get a bargain, given how thin the room is. Nothing in the system tells you to buy. It tells you what things are worth and what they cost, and it records what you decided so it can grade itself later.

What it produces, every single day

Output	Definition
Company file — all ~286 names	Normalised financials, 5–10 year ratio history with trend direction, sector-appropriate analysis, three-to-five independent fair values, a blended fair value with a dispersion band, and a plain-English research note.
The price ladder	Five zones per stock: Strong Accumulate · Accumulate · Fair · Trim · Exit. Derived from fair value and a computed (never guessed) margin of safety. Moves automatically when macro or fundamentals move.
Opportunity ranking	Every investable name ranked by risk-adjusted expected return over the horizon, not by raw upside. Includes what it would cost you to actually build the position.
Regime read	Where the Sri Lankan macro cycle sits, with probability, and what that implies for total exposure and sector tilt.
Portfolio watch	For each holding: has the thesis you wrote down actually changed? Which trigger, if any, has fired?
Decision record	Every decision you take, with the full model state frozen at that moment, so the system can later tell you whether the model was wrong or you were.

The five things that make this work — and they are not the models

#	Component	Why it carries the return
1	Clean point-in-time data	Almost nobody in this market has it. It is unglamorous and it is the moat. Restated financials in a backtest manufacture alpha that does not exist.
2	Cost discipline	Round-trip friction is ~2.24% in fees before spread and impact. A long-horizon, low-turnover design is not a preference here; it is the only design that survives.
3	The macro regime layer	CSE equity is priced as a substitute for Treasury bills. Getting exposure right beats getting selection right, because the market re-rates as a bloc with the rate cycle.
4	Earnings-integrity veto	In a market of family-controlled conglomerates with opaque related-party flows, avoiding the bad half beats picking the good half.
5	Model plurality with visible disagreement	Three fair values that disagree by 40% is information, not noise. The system surfaces the spread and widens the required margin of safety automatically.

THE ONE RULE THAT GOVERNS EVERYTHING ELSE

Deterministic code computes; AI explains. Every ratio, every discounted cash flow, every factor loading, every position size is arithmetic in version-controlled Python with a config hash. The language model reads those numbers and writes the research note, flags the contradictions, triages the announcements and proposes improvements. It never produces a number that lands in a valuation, and it never has write access to the database or to order flow. This separation is what makes the system auditable, and auditability is what makes it improvable.

Part A • Foundations

1 • Purpose — the nine questions

The platform exists to answer nine questions, in this order, for every listed company. Every screen, every table and every stored field must trace back to one of them.

#	Question	Answered by
1	Is this a business worth owning at any price?	Fundamental engine (\$12) + integrity veto (\$14) + sector framework (\$15)
2	Why is it attractive or unattractive?	Pillar sub-scores with trend direction + AI research note (\$44)
3	What is it actually worth?	Valuation engine — 3 to 5 independent anchors, blended (\$16–24)
4	What price should I pay?	Price ladder = fair value less computed margin of safety (\$25–26)
5	What is the expected upside and downside?	Scenario engine: bear / base / bull, plus simulated distribution (\$23)
6	What are the risks and catalysts?	Risk register, event calendar, sector sensitivity matrix (\$33), announcement triage
7	When should I take profit or sell?	Exit framework — five independent triggers (\$28)
8	Has my thesis changed since I bought?	Thesis-drift monitor against the frozen decision record (\$42)
9	Where is the best risk-adjusted opportunity right now?	Opportunity ranking, expected return per unit of residual volatility, net of the cost of building the position (\$40)

2 • Market reality — the constraints you cannot design around

2.1 The cost floor

All-in CSE transaction costs are approximately **1.12% one way** for trades up to LKR 100m — 0.640% brokerage, 0.300% share transaction levy, 0.084% CSE fee, 0.072% SEC cess, 0.024% CDS — which is **~2.24% for a round trip**. Add bid-ask spread and market impact in a thin order book and realistic all-in friction is **3–5% per round trip on mid-caps**.

CONSEQUENCE

Before you have made a single rupee you are 3–5% behind. This single fact kills any design that trades frequently, and it is why the system enforces a minimum expected return floor on every candidate. A model that is right 60% of the time but trades 60 times a year loses money here.

2.2 Structural facts

Fact	Design consequence
~286 listed companies; thin analyst coverage below the S&P SL20	Fundamental mispricing persists for months, not days. A 12–36 month horizon is where the money is.
Market capitalisation ~LKR 7.7 trillion; typical daily turnover LKR 1–3bn, spiking to 8–12bn on crossings	Turnover is roughly 0.02% of market cap per day. Liquidity, not idea generation, is the binding constraint on position size.
Retail and high-net-worth dominated flow; limited institutional arbitrage	Sharp short-term overreaction and slow medium-term information diffusion. Both are exploitable at different horizons.
Extreme sensitivity to Treasury yields — equity treated as a fixed-income substitute	The earnings-yield-minus-364-day-T-bill spread is the single most powerful macro variable in the system. Build the macro layer around it.
Family-controlled conglomerates; opaque related-party flows; revaluation-inflated book values	Earnings-quality screening is unusually high-value. "Hard book" (revaluation reserves stripped) is the primary book value measure.
Large share of 31 March fiscal year ends; filings lag 45–90 days	Point-in-time discipline is not optional. A March-year-end company's audited numbers cannot enter a portfolio formed before end-September.
Meaningful historical delisting and suspension activity	Survivorship-inclusive backtesting. Excluding dead names overstates returns by several points a year.
No commercial factor library exists for Sri Lanka	Building CSE-native size, value, momentum and liquidity factors is itself a durable asset.

CAPACITY — THE HONEST VERSION

This edge works at roughly LKR 5m–200m of deployed capital. It degrades badly above that, because the liquidity that deters institutional money is the same liquidity that caps your own size. Illiquidity is the fence around your field, not a bug. The system must therefore report a *capacity curve*, not just a return.

2.3 State of the market — calibration snapshot, August 2026

Every specification decision below was made against this backdrop. These figures are illustrative of the *current* regime; the system reads them live and never hard-codes them.

Variable	Latest	Reading
Overnight Policy Rate	8.75%	Raised 100bp in May 2026 — first hike since March 2023 — and held at the July review. Next review 30 September 2026.
Headline inflation (CCPI)	6.8% y/y (June 2026)	Above the 5% target, driven by energy and food after the Middle East conflict lifted oil. Expected to return to target only gradually.
Treasury bill yields	~10.1–10.3% across 3m / 6m / 12m	Risen across the curve. This is the hurdle every equity must clear.
Gross official reserves	US\$6.45bn (end-June 2026)	Down from US\$6.8bn in April amid debt servicing. Drain, not accumulation.
IMF EFF	Combined 5th & 6th reviews completed 27 May 2026	~US\$695m disbursed; cumulative ~US\$2.4bn of the SDR2.286bn facility. Final year of the programme.
2026 GDP growth	Cut to ~3.0%	Down from 2025's ~5%. Cyclone Ditwah reconstruction plus oil shock plus weaker tourism.
ASPI	~21,270 (6 Aug 2026)	Peaked near 22,280 in early June, then fell ~800 points immediately after the 26 May rate hike. A textbook demonstration of the T-bill-substitute thesis.
S&P SL20	~5,970	Large caps holding up marginally better than the broad market.
Foreign flow	Persistent net outflow	Domestic institutional and HNW buying is absorbing it. Turnover concentrated in crossings — headline turnover overstates real liquidity.

REGIME READ AS AT THIS SNAPSHOT

Transition, tilting toward Risk-Off. Rising yields, inflation above target, currency pressure, reserve drain, growth downgrade, oil shock — against a supportive reform anchor (IMF programme on track) and reconstruction spending. This is a "quality, low leverage, defensive, reduced gross exposure" configuration, not a "buy the small-cap beta" one. The system should be widening margins of safety and raising the discount rate right now, and the specification below makes that happen automatically rather than by opinion.

3 • Edge thesis — why this can work here specifically

Structural feature	Exploitable consequence	Harvested by
Thin coverage, small universe	Mispricing persists for months	Fundamental + valuation engines
Retail-dominated flow	Overreaction short-term, slow diffusion medium-term	Timing engine (\$37)
Equity-as-bond-substitute	Market-wide re-ratings forecastable with 1–3 month lead	Macro engine (\$29–33)
Opaque related-party structures	Avoiding the bad half is high-value	Integrity veto (\$14)
Illiquidity	Deters institutional capital, so the mispricing is not arbitraged away	Universe gates + sizing caps
No factor library exists	Building CSE-native factors is itself an edge	Factor library (\$35)
Revaluation-inflated book value is endemic	"Value" screens on reported book are systematically wrong	Hard-book adjustment (\$22)

4 • Design laws

Seven rules. A change request that violates one of these is rejected regardless of how good the backtest looks.

#	Law	What it means in practice
1	Code computes, AI explains	Valuations, ratios, indicators, sizes: deterministic Python, version-controlled, hash-stamped. The LLM narrates and proposes. No AI-generated number enters a calculation.
2	Point-in-time or nothing	Every query filters on <code>first_available_date</code> . Restated statements are inserted as new versions, never overwritten.
3	Never invent data	Missing is displayed as missing. Estimated, derived and forecast values carry a visible provenance chip. Silence is a lie in this system.
4	Nothing is a black box	Every score expands to its inputs, and every input to its source and timestamp. You cannot sensibly override what you cannot inspect.
5	Surface disagreement, don't average it away	When models conflict the UI makes it impossible to miss, and the required margin of safety widens automatically. Consensus is cheap; conflict is where judgement earns its keep.
6	The system presents; the human decides	There is no BUY button. There is an evidence panel and a decision-capture form. This is a deliberate design constraint, not a limitation.
7	Modularity	Any engine can be replaced without rebuilding the rest. Contracts between layers are typed and tested. Data flows up; constraints flow down.

Part B • Data Layer

5 • Sources

Domain	Source	Frequency	Method
Prices, volume, turnover, trades, order book	cse.lk JSON endpoints; broker EOD files	5-minute during market hours; EOD snapshot	Apify actor → schema-validated loader
Company financials (quarterly + annual)	CSE company disclosures (PDF), annual reports	Quarterly / annual	PDF table extraction → LLM-assisted line-item mapping → mandatory human confirm queue
Notes to accounts, segment data, related-party notes	Annual report PDFs	Annual	LLM extraction into structured fields; every field carries a page-anchor citation
Corporate actions (splits, rights, bonus, dividends)	CSE announcements	Event-driven	Scrape + mandatory human confirm — this is the highest-consequence data in the system
Shareholding, free float, top-20 holders	Quarterly disclosures, annual reports	Quarterly	Scrape
Foreign net flow, market breadth	CSE daily market summary	Daily	Scrape
Policy rate, AWPLR, T-bill and bond yields, M2b, private credit, reserves	CBSL	Daily / weekly / monthly	API + scrape, release-calendar driven
CCPI headline & core, NCPI	DCS / CBSL	Monthly	Scrape
LKR/USD, NEER, REER	CBSL	Daily	Scrape
Tourist arrivals, remittances, trade balance, sector credit	CBSL / SLTDA	Monthly	Scrape
Sovereign spread, rating actions, IMF review milestones, budget documents	IMF, rating agencies, Treasury, news	Event-driven	Scrape + manual event log
National project pipeline (ports, power, roads, FDI announcements)	BOI, Treasury, ministry releases, credible press	Event-driven	LLM extraction → structured project register (\$34) → human confirm
Brent, global EM risk proxies, US policy rate, commodity prices (tea, rubber, palm oil)	Public feeds	Daily	API
Sector-level valuation and fundamental reference data	Damodaran datasets (industry betas, margins, reinvestment rates, ERP by country)	Semi-annual	Download and version — used as sanity bands, never as a substitute for local estimation

THE SINGLE BIGGEST OPERATIONAL FRAGILITY

The cse.lk JSON endpoints are reverse-engineered rather than officially documented, and they require no authentication but do warrant sensible self-imposed rate limiting. Treat them as a dependency that will break without notice: schema validation on every response, contract tests in CI, alerting on *shape change* rather than only on HTTP error, conservative pacing (≥2s between calls), exponential backoff, a circuit breaker, an identifying user-agent, and never parallel hammering. Identify a second source now — a broker EOD file is the cheapest option — because a single unofficial endpoint is a single point of failure. If usage ever scales beyond personal, move to a licensed feed.

6 • Point-in-time discipline

Every fundamental record carries three dates. This is the non-negotiable rule of the data layer.

period_end	the fiscal period the number describes
first_available_date	the date the market could first have seen it (release date)
ingest_date	when this system stored it

All models query on `first_available_date <= t`, never `period_end <= t`. Financial statements are immutable and versioned: when a company restates, you *insert a new row*. You never update the old one. Restated data leaking into a backtest is the single most common source of alpha that does not exist.

CSE-specific timing. Default assumption: 3-month reporting lag on quarterlies, 6-month on audited annuals, verified per company and stored on the security record. Given the concentration of 31 March fiscal year ends, the annual factor formation date is **30 September**, not the Fama-French convention of June — this guarantees a six-month lag after the dominant year end and eliminates look-ahead.

7 • Corporate actions — where CSE backtests silently die

A single unadjusted 1:1 bonus issue creates a fake ~50% return that a momentum model will read as a signal. Requirements:

- A **total-return adjustment factor series per ticker**, applied cumulatively backwards across the entire price history.
- Rights issues adjusted using the **theoretical ex-rights price (TERP)**, not simple dividend adjustment. $TERP = (N \times \text{cum-rights price} + S \times \text{subscription price}) \div (N + S)$.
- A **nightly reconciliation test**: recompute each stock's total return from adjusted prices, and independently from raw prices plus declared actions. Any mismatch >0.5% raises a data alert and **quarantines that ticker from every model** until a human resolves it.
- Delisted, suspended and defaulted companies remain in the database and in every backtest until their delisting date, carrying a terminal return (recovery value, or -100%).

8 • Provenance and quality tiering

Every number the platform displays carries a provenance tier. This is rendered in the interface as a small chip, and it is queryable. Nothing is ever shown without one.

Tier	Label	Definition	Rules
R	Reported	Lifted directly from an audited or filed statement	Must carry source document URL, page anchor, and <code>first_available_date</code> . Highest trust.
D	Derived	Deterministic arithmetic on Reported inputs (e.g. ROIC, EV/EBIT)	Formula version hash stored. Fully reproducible. Inherits the worst provenance of its inputs.
N	Normalised	Reported data adjusted for one-offs, accounting policy change, restatement, or revaluation reserves	Adjustment must be itemised and expandable in the UI. Original value always retained alongside.
E	Estimated	Statistically filled from peers, sector medians or interpolation	Method named. Confidence band shown. Excluded from any hard gate or veto decision.
F	Forecast	Forward projections from the model — revenue growth, margins, terminal values	Assumption set must be visible and editable by the user. Scenario-tagged.
A	AI-assisted	Extracted or inferred by a language model — e.g. a segment breakdown parsed from a PDF note	Must show the source snippet. Cannot enter a valuation until human-confirmed and promoted to Reported. Displayed with a distinct chip.
—	Unavailable	Genuinely not obtainable	Rendered as "Data unavailable" with the reason. Never silently zero, never quietly imputed, never omitted from the layout.

DOWNSTREAM RULE

A composite score inherits the *weakest* provenance among its material inputs, and the UI shows a data-completeness percentage on every company card. A stock scoring 88/100 on 54% data completeness is a fundamentally different object from one scoring 84/100 on 97%, and the ranking must let you see that difference before you act on it.

Freshness. Every record carries `fetches_at`. The UI shows a stale-data banner when the last successful fetch is older than the expected interval, and **models refuse to emit new signals on stale inputs**. A failed fetch must never silently produce a stale signal — fail loudly, degrade visibly, and label yesterday's number clearly as yesterday's.

9 • Core schema

```
securities      (ticker, isin, name, gics_sector, cse_sector, archetype,
                 listing_date, delisting_date, board, fiscal_year_end,
                 reporting_lag_days, currency_exposure_profile)
prices_daily    (ticker, date, open, high, low, close, vwap, volume,
                 turnover, trades, adj_factor, fetched_at)
prices_intraday (ticker, ts, last, best_bid, best_ask, depth_bid, depth_ask)
corporate_actions (ticker, ex_date, type, ratio, cash_amount, terp,
                  source_url, confirmed_by, confirmed_at)
fundamentals    (ticker, period_end, period_type, first_available_date,
                 version, statement_line, value, currency, provenance_tier,
                 restated_flag, source_url, source_page)
segments        (ticker, period_end, segment_name, revenue, ebit, assets,
                 provenance_tier, source_page)
float_data       (ticker, as_of, shares_issued, public_float_pct,
                 top20_pct, controlling_holder)
macro_series    (series_id, obs_date, first_available_date, value, source)
projects         (project_id, name, sector, capex_lkr, status, timeline,
                 sponsors, affected_tickers[], source_url, confirmed_by)
events          (event_id, date, type, description, affected_scope,
                 sentiment, horizon_tag)
factor_returns   (date, mkt_rf, smb, hml, hml_hard, mom, liq, rf)
valuations       (ticker, as_of, method, scenario, fair_value, inputs_json,
                 config_hash, provenance_tier)
model_outputs    (ticker, as_of, model_id, score, subscores_json,
                 diagnostics_json, model_version, config_hash)
decisions        (see §45 — full model state frozen at decision time)
outcomes         (see §45)
positions / trades / alerts / journal / data_alerts
```

Part C • Coverage

10 • Analyse everything, invest in a subset

You asked that *all* stocks be analysed. The original architecture gated the universe down to ~90–140 names before any model ran. Both are right, and the resolution is to separate **analysis coverage** from **investment eligibility**.

RESOLUTION

Every one of the ~286 listed companies gets a full company file, a valuation and a research note. The liquidity and structural gates no longer decide who gets analysed — they decide who is *eligible for capital*, and they do so as a visible, explained label rather than an invisible filter. Nothing is hidden from you; things are labelled. This costs almost nothing computationally and it means a name that becomes liquid, or recovers from a watch-list, is already in your history rather than appearing from nowhere.

11 • Coverage tiers

Tier	Expected count	Definition	What the system does
Core	~90–140	Passes liquidity, structural and integrity gates	Full analysis, valuation, scoring, ranking, sizing, alerting. Eligible for capital.
Watch	~60–100	Fails one gate, usually liquidity, but is otherwise sound	Full analysis and valuation. Ranked in a separate list. Alerts if the failing gate clears. Not sized.
Excluded	~30–60	Fails a hard integrity or structural gate — going-concern qualification, Beneish above threshold, suspension, watch list	Analysed and valued, but shown with the exclusion reason prominent. Never enters a buy list at any score.
Insufficient	~10–30	Fewer than 8 quarters of usable history, or data completeness below 40%	Company file with whatever exists, no fair value published. Shown as "cannot value — here is why".

11.1 Gate definitions

Gate 1 — Liquidity (soft; sets eligibility and caps size). Thresholds are parameters that scale with your capital, not constants.

Test	Threshold	Rationale
Median daily turnover, 60 sessions	≥ LKR 2.0m	Below this you cannot enter or exit at a rational price
Days traded, last 60 sessions	≥ 45 (75%)	Non-trading breaks every return calculation and biases beta downward
Amihud illiquidity, 60d	Below universe 80th percentile	Direct impact-cost proxy
Your max position ÷ 20d ADV	≤ 15%	Defines the position cap, not merely eligibility

Gate 2 — Structural (hard). Free float ≥15% of shares issued · not on the Watch List or under trading suspension · listed ≥12 months (no IPO-pop chasing) · market cap ≥ LKR 1.0bn · at least 8 quarters of continuous financial history.

Gate 3 — Integrity (hard veto, never a scored input). Qualified audit opinion or emphasis-of-matter on going concern · auditor change and CFO departure within the same 12 months · Beneish M-Score > -1.78 · related-party revenue or receivables >30% of total. Any one of these excludes the name from capital regardless of how cheap it looks. In a market with these ownership structures, this veto is worth more than any refinement to the valuation models.

WHY THE VETO MUST NEVER BE A WEIGHTED SCORE

If integrity is a scored input, a sufficiently attractive valuation will always outvote it — which is precisely the trade that destroys capital in family-controlled markets. Make it a gate and the question never arises.

Part D • Company Analysis

12 • Fundamental engine

Computed for every company, every quarter, over the longest available history (target 10 years, minimum 8 quarters). All ratios are computed within sector as cross-sectional percentiles, winsorised at 1%/99%, so that a 12% ROE means something different in banking than in plantations.

Profitability

- ROE, ROA, ROIC
- ROIC – WACC spread (*the real question*)
- Gross margin, EBITDA margin, EBIT margin, net margin
- Gross profitability (gross profit ÷ total assets) — Novy-Marx; robust where earnings are noisy
- Operating cash flow margin
- Cash conversion: CFO ÷ net income, 3-year average

Growth

- Revenue, EBITDA, EBIT, net income, EPS, book value, FCF, dividend CAGR
- Measured over 3, 5 and 10 years where data exists
- **Fundamental growth** = reinvestment rate × ROIC — the growth the business can actually fund, as distinct from the growth it happened to report
- Growth *quality*: is it volume, price, acquisition, or currency?

Financial strength

- Debt/equity, net debt/EBITDA, interest coverage
- Current and quick ratio, cash/debt
- Debt maturity profile; refinancing cliff = short-term debt ÷ (cash + CFO)
- Altman Z"-Score (emerging-market variant)
- **FX exposure**: unhedged foreign-currency debt ÷ total debt — critical here; a 10% LKR depreciation has repeatedly wiped out the equity of importers and USD borrowers

Efficiency

- Asset turnover, inventory turnover
- Receivable days, payable days, cash conversion cycle
- Working capital as % of revenue, and its trend
- Capex ÷ depreciation — is the asset base being maintained or harvested?

Shareholder returns

- Dividend yield, payout ratio, dividend CAGR
- Dividend cover from FCF, not from earnings
- Retained earnings growth and what it earned
- ROE consistency: inverse standard deviation over 5 years
- Share count trend — dilution from rights and share-based payment

Composite accounting scores

- **Piotroski F-Score** (0–9), displayed as its own number — unusually high signal on cheap small caps
- **Altman Z"** distress score
- **Beneish M-Score** manipulation score
- **Sloan accrual ratio** = (net income – CFO) ÷ average total assets

13 • Trend detection — the point of the whole exercise

A ratio is a fact; a trajectory is an insight. "ROE = 18%" is far less useful than "ROE moved 11% → 14% → 16% → 18% over four years, driven by margin rather than leverage." The engine must therefore emit, for every material metric:

Output	Definition
Direction	Sign and significance of the slope over 3, 5 and 10 years (Mann-Kendall trend test to avoid reading noise as trend)
Acceleration	Second derivative — is the improvement speeding up or fading?
Consistency	Coefficient of variation; how many of the last N periods moved the same way
Percentile shift	Movement in the company's rank within its sector, which strips out sector-wide effects
Decomposition	DuPont for ROE (margin × turnover × leverage); revenue into price vs volume vs mix vs FX where segment data allows. This tells you <i>why</i> the trend exists, which is the only version that survives contact with a changing environment.
Break detection	Structural break in the series — an accounting policy change, an acquisition, or a genuine inflection. Flagged, not smoothed over.

14 • Earnings integrity

Twenty percent of the fundamental composite, **and a veto**. Signals:

- **Beneish M-Score** — eight-variable manipulation probability model
- **Sloan accrual ratio** — high accruals relative to cash flow predict earnings reversal
- **Related-party revenue and receivables** as % of total, and the trend — the central risk in a family-conglomerate market
- **Days sales outstanding trend** over three years — the earliest warning of channel stuffing or a collection problem
- **Auditor tier** and any opinion modification, plus auditor changes
- **Director and insider net buying** over 12 months
- **Revaluation dependence**: what proportion of book equity and of reported profit comes from asset revaluation and fair-value gains rather than operations
- **Cash conversion divergence**: profit rising while CFO falls, for more than two consecutive periods

15 • Sector frameworks — the right lens per business

Routing is automatic, by GICS sector mapped to a CSE-native **archetype** stored on the security record. The archetype determines which ratios are computed, which are suppressed as meaningless, which valuation models run, and how the anchors are weighted.

Archetype	Metrics that matter	Metrics that are meaningless here	Primary valuation
Banking	NIM and trend, CASA ratio, cost-to-income, Stage 3 loans, provision cover, credit cost normalisation, Tier 1 CAR headroom, loan and deposit growth, ROE, ROA	EV/EBIT, EV/EBITDA, ROIC, free cash flow, net debt	Justified P/TBV from ROE · residual income · multi-stage DDM
Non-bank finance / leasing	Yield on advances, cost of funds, spread, NPL ratio, portfolio growth, gearing, liquidity coverage, funding concentration	Same as banking	Justified P/B · residual income · DDM
Insurance	GWP growth, combined ratio (general), embedded value and VNB (life), investment yield, solvency ratio, persistency	EV/EBITDA, ROIC	Embedded value + VNB multiple (life) · P/B and ROE (general)
Diversified holdings	Segment EBIT and its trend, NAV, holding-company discount vs its own history, capital allocation record, intra-group receivables, cash at holdco	Consolidated margins (they mix unlike businesses and mean nothing)	Sum-of-the-parts · segment-level EV/EBIT · NAV gap
Manufacturing / industrials	Capacity utilisation, input cost pass-through lag, ROIC, export share, FX exposure both ways, capex ÷ depreciation, order book	Dividend yield as a primary anchor	FCFF DCF · EV/EBIT vs own and sector history
Consumer / retail	Volume vs price growth split, pricing power (margin stability through inflation), same-store growth, working capital cycle, distribution reach	Asset-based valuation	FCFF DCF · EV/EBITDA · P/E vs own history
Plantations	Tea / rubber / palm oil realised prices vs auction average, yield per hectare, cost per kilo, labour cost per hectare, mature vs immature hectarage, replanting rate, weather exposure	Reported book value (revaluation-inflated), P/E in a commodity trough	Normalised mid-cycle earnings × through-cycle multiple · hard NAV per hectare
Hotels / tourism	Occupancy, ADR, RevPAR, arrivals by source market, room inventory pipeline, USD revenue share, net debt/EBITDA, refurbishment cycle	Trailing P/E across a cycle; reported book equity	Normalised EBITDA × EV/EBITDA · replacement cost per key · hard-book NAV
Telecom / technology	ARPU, subscriber growth, data mix, capex intensity, spectrum obligations, USD capex exposure, EBITDA margin	Book value	FCFF DCF · EV/EBITDA
Construction / materials	Order book and cover ratio, government receivable days, input costs (cement, steel, bitumen), capacity utilisation, exposure to the state as counterparty	Dividend stability	FCFF DCF with explicit order-book fade · EV/EBIT
Power & energy	Plant load factor, tariff structure and regulatory reset dates, PPA tenor, fuel pass-through mechanism, USD debt	Growth multiples	Contracted-cashflow DCF · regulated asset base
Property	Occupancy, rental yield, development pipeline, pre-sales, land bank at cost vs market, gearing	Reported P/E; consolidated ROIC	NAV with independent land marks · development DCF

THE RULE THIS TABLE ENCODES

Do not apply an industrial DCF to a bank. A bank's debt is its raw material, not its financing; free cash flow to the firm is not a meaningful concept for it. Get this wrong and the model will produce a confident, precise, entirely fictional number. The archetype router exists specifically to make that error impossible.

Part E • Valuation Engine

16 • Model router — choosing the right instrument

Following Damodaran's model-choice logic, the router selects models per company from the answers to five questions, all of which the system can answer from stored data.

Question	Routing consequence
Are cash flows positive and reasonably predictable?	No → DCF is downweighted or suppressed; normalised-earnings and asset-based anchors take over
Is this a financial firm?	Yes → equity-side models only (DDM, FCFE, residual income, justified P/B). Firm-side models suppressed.
Are dividends a meaningful proxy for capacity to pay?	Yes and payout is stable → DDM weight rises. No → FCFE.
Is the business cyclical or commodity-linked?	Yes → all earnings normalised to a mid-cycle average before any multiple is applied
Is it a holding company with separable segments?	Yes → sum-of-the-parts becomes the primary anchor
Is the company distressed, or is equity effectively an option on recovery?	Yes → option-based equity valuation runs as a supplementary anchor, flagged as such

The router's decision, and the reason for it, is displayed on the stock page. If a model was suppressed, the user sees which one and why.

17 • Discount rates — the hardest number in Sri Lankan valuation

The CFA Institute's practitioner survey is blunt about this: evaluating the discount rate is more than an estimation problem, it is a conceptual one, and most of the finance literature simply assumes a Treasury curve is the correct riskless rate. In a country that defaulted in 2022 and restructured 98% of its external debt, that assumption is not available. The system therefore constructs the cost of equity explicitly and shows every component.

17.1 The risk-free rate

HOUSE CONVENTION — AND THE DOUBLE-COUNT TRAP

The system values in **LKR nominal**, discounting LKR cash flows at an LKR rate. Two routes exist and they must not be mixed:

Route A (default). $Rf_LKR = 364\text{-day T-bill yield}$ (currently ~10.2%). Domestic rupee government paper has not been restructured and is the relevant opportunity cost for a domestic rupee investor. The equity risk premium added on top is then a **mature-market ERP plus a reduced country risk premium**, because the local nominal rate already embeds Sri Lankan inflation and a substantial part of sovereign risk. Adding a full Damodaran country risk premium on top of a 10.2% local risk-free is **double-counting**, and it will value every company on the exchange at roughly zero.

Route B (cross-check). $Rf_LKR = \text{local government bond yield} - \text{sovereign default spread}$, then add the full country-inclusive ERP. Run this quarterly as a sanity check on Route A. If the two routes diverge by more than ~200bp in the implied cost of equity, the macro engine raises a flag for human review.

Whichever route, the resulting Ke is displayed with its components broken out, and the implied ERP is reverse-engineered from current ASPI earnings yield as a third reference point.

17.2 Cost of equity

$$Ke = Rf_LKR + \beta_adjusted \times ERP_effective + size_premium + illiquidity_premium$$

Rf_LKR	364-day T-bill primary auction yield, weekly
$\beta_adjusted$	Dimson-corrected beta (see §35.2), Blume-adjusted toward 1.0, blended with a bottom-up sector beta unlevered/relevered at the company's own D/E. Bottom-up dominates where the stock trades fewer than 45 of 60 sessions.
$ERP_effective$	Mature-market ERP + residual country premium under Route A; reviewed semi-annually against Damodaran's country dataset and against the ASPI implied premium.
$size_premium$	0 to ~2.5%, banded by free-float market cap decile
$illiquidity_premium$	0 to ~3.0%, mapped from the Amihud percentile. Explicit, not buried. If you cannot sell it, it is worth less to you.

Cost of debt = $AWPLR + \text{company-specific spread inferred from interest expense} \div \text{average debt}$, cross-checked against coverage-implied synthetic rating. **WACC** uses market-value weights, with tax at the statutory corporate rate adjusted for observed effective rate.

REGIME LINKAGE — THIS IS WHERE THE MACRO LAYER PAYS FOR ITSELF

The equity risk premium and the risk-free rate are both regime-conditional inputs supplied by the macro engine (§31). When the regime flips toward Risk-Off, Ke rises, and **every fair value in the system falls automatically, overnight, without anyone forming an opinion**. The macro view is not decoration on a dashboard; it lands directly in the buy price of every stock.

18 • Discounted cash flow

18.1 Structure

Three-stage FCFF model by default for operating companies; FCFE for financials and for companies where leverage is being actively changed.

```

FCFF = EBIT × (1 – effective tax rate)
      + depreciation & amortisation
      – capital expenditure
      – change in non-cash working capital

Stage 1  Years 1–5    explicit forecast, line by line
Stage 2  Years 6–10  linear fade of growth and margin toward stable state
Stage 3  Terminal    stable growth, capped at long-run nominal GDP growth

Terminal value = FCFFn+1 ÷ (WACC – g)
with the discipline: g ≤ long-run nominal GDP growth AND
                    reinvestment rateterminal = g ÷ ROICterminal
(a company cannot grow forever without reinvesting for it)

Equity value = Σ FCFFt ÷ (1+WACC)t + PV(terminal)
              + cash and non-operating assets
              – total debt – minority interest – pension deficit
              + diluted shares outstanding

```

18.2 Assumption sourcing — never a free parameter

Input	Derived from
Revenue growth Y1–2	Trailing 3-year CAGR, adjusted by sector macro sensitivity (§33) and any confirmed project in the register (§34). Where quarterly guidance or an order book exists, it overrides.
Revenue growth Y3–5	Fade toward sector median growth × macro regime multiplier
Operating margin	5-year average, adjusted for confirmed structural change; fades toward sector median unless ROIC–WACC spread evidences durable advantage
Tax rate	Effective rate trailing 3 years, converging to statutory by Y5
Capex	Trailing capex ÷ revenue, floored at depreciation ÷ revenue in the terminal year
Working capital	Trailing non-cash WC ÷ revenue, held constant unless a trend test is significant
Terminal growth	MIN(long-run nominal GDP growth, sector terminal growth, R _f –LKR). Never exceeds the risk-free rate — a company growing faster than the risk-free rate forever is worth infinity, which is a sure sign of a modelling error.

Every one of these is **editable in the interface**, and every edit is stored as a user-scenario alongside the system base case, so the two are never confused.

19 • Dividend and residual income models

19.1 Gordon growth

$$V_0 = D_1 \div (K_e - g)$$

Applied only where payout has been stable for five years, growth is below K_e , and the business is mature. Used mainly as a floor check on mature financials and utilities, never as a sole anchor.

19.2 Multi-stage DDM

Two- and three-stage variants for banks and finance companies with a visible growth-to-maturity path. Dividend capacity is capped by **sustainable payout** = $1 - (g \div ROE)$, so a bank cannot be credited with paying dividends it would need to retain to fund its own loan growth.

19.3 Residual income — the right model for CSE banks

$$\begin{aligned}
V_0 = & \text{Book value of equity} \\
& + \sum [(ROE_t - K_e) \times \text{Book}_{t-1}] \div (1 + K_e)^t \\
& + \text{terminal residual income}
\end{aligned}$$

$$\text{Justified P/B} = (ROE - g) \div (K_e - g)$$

WHY THIS MATTERS DISPROPORTIONATELY HERE

Banks are a large share of the S&P SL20. Residual income anchors most of the value in *book equity*, which is observed and audited, rather than in a terminal value fifteen years away, which is invented. In a market where forecasting is genuinely hard, moving value from the invented part of the model to the observed part is a direct improvement in reliability. The **ROE-versus-P/B cross-sectional regression** across all listed banks — plotting each bank's P/B against its ROE and reading off the residual — is the single cleanest relative-value signal available in Sri Lankan equities.

20 • Relative valuation — done properly

Practitioners overwhelmingly use multiples; the CFA survey puts market multiples ahead of discounted cash flow in frequency of use. But multiples mislead in three known ways: comparable firms are hard to select, different multiples suggest conflicting conclusions, and not all multiples are meaningful in all contexts. The system addresses each explicitly.

20.1 Three comparison frames, always shown together

Frame	Computation	What it tells you
Own history	Current multiple vs its own 5- and 10-year median and interquartile range, with the percentile stated	Is this cheap <i>for this company</i> ? Immune to the "whole peer group is overvalued" trap.
Sector peers	Premium/discount to sector median and to a matched peer set (similar size, leverage, growth)	Is this cheap <i>relative to alternatives</i> you could own instead?
Market	Vs ASPI and S&P SL20 aggregate multiples, and vs the earnings-yield-minus-T-bill spread	Is the whole market cheap or expensive against the risk-free alternative?

20.2 Justified multiples — the fix for naive comparables

Rather than asking "is this P/E lower than the sector median?", the engine asks "**what P/E does this company's own fundamentals justify, and is it trading below that?**" This is the Damodaran anatomy-of-a-multiple approach and it removes most of the comparability problem.

```

Justified P/E = payout × (1 + g) ÷ (Ke - g)
Justified P/B = (ROE - g) ÷ (Ke - g)
Justified EV/EBIT = (1 - tax) × (1 - g/ROIC) ÷ (WACC - g)
Justified P/S = net margin × payout × (1 + g) ÷ (Ke - g)

```

In parallel, a **cross-sectional regression** across the whole exchange:

```
P/B_i = a + b·ROE_i + c·growth_i + d·leverage_i + e·payout_i + ε_i
```

The residual ϵ is the mispricing estimate. A company with a low P/B *and* a low ROE is not cheap — it is correctly priced. Only a negative residual is cheap. This regression is re-estimated monthly, run separately by archetype, and displayed as a scatter with the company's position marked, because a picture of where you sit relative to the fitted line communicates in two seconds what a table communicates in two minutes.

20.3 Multiples used, and their normalisation

Multiple	Numerator / denominator discipline	Applicable archetypes
P/E, forward P/E	Normalised EPS — 3-year average for cyclicals; one-offs stripped; dilution fully counted	All except deeply cyclical in a trough
PEG	Only against <i>fundamental</i> growth (reinvestment × ROIC), never against a naive historical CAGR	Growth companies with positive, stable earnings
P/B and P/hardB	Hard book strips revaluation reserves on land and buildings. Hard book is primary.	Financials, hotels, plantations, property
EV/EBIT, EV/EBITDA	EV includes minorities, leases and pension deficits; EBIT normalised over 3 years	Operating companies
EV/Sales	Only where margins are temporarily depressed but structurally recoverable	Turnarounds, early-stage
FCF yield (FCF ÷ EV)	FCF after maintenance capex; 3-year average	All operating companies
Dividend yield	Shown against the 364-day T-bill yield, always — the honest comparison for a Sri Lankan investor	Mature payers
EV per unit	Per hotel key, per hectare, per MW, per subscriber	Hotels, plantations, power, telecom

21 • Sum-of-the-parts

The CSE-native valuation method, because the largest names are conglomerates.

```

SOTP equity value =
  Σ (listed subsidiaries at market value × ownership %)
+ Σ (unlisted subsidiaries at appropriate sector EV/EBITDA or EV/EBIT)
+ investment property at hard book or independent mark
+ associates at equity-method carrying value or market where listed
+ net cash at the holding company
- holdco net debt
- holding-company discount
÷ shares outstanding

```

THE HOLDING-COMPANY DISCOUNT MUST BE EARNED, NOT ASSUMED

Do not apply a flat 20%. Calibrate it to **that company's own historical average discount to its own NAV**, then adjust for three observable factors: capital-allocation record (has reinvested capital earned above its cost?), intra-group related-party flow intensity, and free float. Typical range 15–35%. Show the calibration. A conglomerate whose discount has widened from 18% to 34% without a deterioration in any segment is a specific, testable opportunity — and that is exactly the kind of thing this market throws up and nobody screens for.

The interface renders SOTP as a **waterfall**: each segment's contribution, then the deductions, then the discount, ending at value per share against current price. Every bar expands to its own valuation.

22 • Asset-based and NAV valuation

Necessary for plantations, property and hotels — and dangerous, because reported book value in these sectors is inflated by property revaluation. Rules:

- **Hard book** is primary: revaluation reserves on land and buildings stripped from equity. Both figures are shown side by side, never one alone.
- Land is marked at an independent per-acre or per-perch reference where obtainable, with the reference and its date shown. Where not obtainable, it stays at cost and is labelled — .
- Replacement-cost cross-check for hotels: EV per key against recent build cost per key.
- Plantations: hard NAV per planted hectare, split mature and immature, cross-checked against recent estate transactions.
- Liquidation value as an absolute floor for distressed names — a genuine margin-of-safety anchor, not a target.

23 • Scenarios and simulation

A single fair value is false precision. Every DCF and every SOTP runs three deterministic scenarios plus a distribution.

Scenario	Construction	Interpretation
Bear	Growth at 25th percentile of its own history, margin at 25th percentile, WACC +150bp, terminal growth -100bp, plus the archetype's specific stress (FX shock for importers, occupancy shock for hotels, commodity trough for plantations, credit-cost spike for banks)	Your downside. The number that should drive position size.
Base	Assumptions as derived in \$18.2	The anchor.
Bull	Growth and margin at 75th percentile, WACC -100bp, plus any confirmed project in the register reaching completion on schedule	What you are being paid to wait for.

Monte Carlo overlay. 10,000 draws over the four inputs the value is most sensitive to, with distributions fitted to each company's own historical variability rather than assumed normal. Output: a fair-value distribution with 10th, 25th, 50th, 75th and 90th percentiles, and — the number that actually matters — **P(fair value > current price)**.

Sensitivity tornado. Always displayed. Which single assumption moves the valuation most? As one practitioner in the CFA study observed, if relatively small changes in a single factor have a big impact on valuation, that itself raises a red flag. The tornado chart makes that flag visible without anyone having to look for it.

Reverse DCF. Alongside every forward valuation, the engine solves for what the *current market price* implies about growth and margin, and states it in plain language: "at LKR 78.50 the market is assuming 11% revenue growth for a decade and margins holding at 14% — this company has exceeded 11% growth in three of the last ten years." Reverse DCF is frequently the most persuasive single output in the entire platform, because it converts a valuation debate into a factual question about the company's own track record.

24 • Triangulation — from many values to one range

Three to five anchors per company, blended by archetype. The weights are defaults; they are configurable and version-controlled.

Archetype	Intrinsic (DCF / RI / DDM)	Asset / SOTP	Relative (justified multiple)	Note
Operating co., stable earnings	0.45	0.15	0.40	DCF-led
Conglomerate / holding	0.20	0.55	0.25	SOTP-led
Bank / finance	0.40 (RI + DDM)	0.35 (justified P/TBV)	0.25	Equity-side only
Cyclical (hotels, plantations, construction)	0.30	0.35	0.35 (normalised)	All earnings mid-cycle
Property	0.25	0.55	0.20	NAV-led
Insurance	0.45 (EV + VNB)	0.35	0.20	Embedded-value led

$$FV_blended = \sum (w_i \times FV_i)$$
$$dispersion = (max\ FV_i - min\ FV_i) \div mean\ FV_i$$

DISPERSION IS A SIGNAL, NOT AN INCONVENIENCE

When three independent methods land within 8% of each other, you understand the business. When they span 45%, you do not — and the honest response is not to average them into a confident-looking number, but to **demand a much larger discount before buying**. Dispersion therefore feeds directly into the margin-of-safety calculation in \$25. This single mechanism converts model uncertainty into a price, automatically, which is exactly what a valuation system should do with uncertainty and almost none of them do.

Part F • Price Engine

This is the component that turns analysis into an executable number. Everything before it is research; everything after it is portfolio management. It runs in four stages.

25 • Margin of safety — computed, never habitual

A flat "20% margin of safety" is a superstition. The required discount should be a function of how much you actually know about the company and how hostile the environment is. It is the sum of five components, each of which is displayed.

Component	Rule	Range	Rationale
Base	Fixed	10%	Covers round-trip friction plus a minimum required edge
Dispersion	$(\text{max FV} - \text{min FV}) \div \text{mean FV} \times 0.5$	0-15%	The more your anchors disagree, the less you know
Liquidity	Amihud percentile mapped: top quartile → 0%, bottom quartile → 10%	0-10%	You will pay to get out. Price it now.
Regime	$\text{Risk-On } 0\% \cdot \text{Transition } +5\% \cdot \text{Risk-Off } +12\%$	0-12%	Supplied by the macro engine. Automatic, not discretionary.
Quality / integrity	$(70 - \text{integrity_score}) \div 4$, floored at 0	0-8%	Pay less for opaque accounts
Data completeness	$(90 - \text{completeness_}\%) \div 5$, floored at 0	0-10%	NEW You cannot value what you cannot see

MoS_total = base + dispersion + liquidity + regime + quality + completeness
→ bounded 10% to 55%

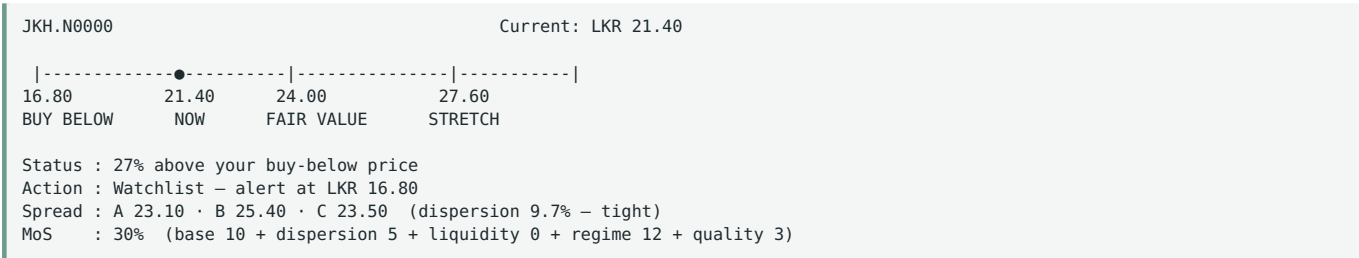
READ A HIGH NUMBER CORRECTLY
A 48% required margin of safety is not the model being pessimistic about the company. It is the model telling you it does not understand this company well enough to pay up. That is a statement about your information, not about the business — and it is far more useful than a confident point estimate would have been.

26 • The price ladder

Five zones, derived, auditable, and they move when the inputs move.

Zone	Formula	Meaning
Strong accumulate	$< \text{FV} \times (1 - \text{MoS} - 0.08)$	Materially below your required entry. Size toward the upper end of the tier cap.
Accumulate	$\text{FV} \times (1 - \text{MoS} - 0.08) \text{ to } \text{FV} \times (1 - \text{MoS})$	Your limit-price band. This is the number you asked for.
Fair	$\text{FV} \times (1 - \text{MoS}) \text{ to } \text{FV}$	Reasonably priced. Hold; do not add.
Trim	$\text{FV to } \text{FV} \times 1.15$	Begin scaling out. Continue only while momentum and thesis remain intact.
Exit	$> \text{FV} \times 1.15$	Valuation stretched. Redeploy — capital has an opportunity cost.

Rendered as a single horizontal bar with the current market price marked on it, and the gap stated in plain language. Illustrative structure only — populated live:



27 • Execution reality — the order book, not the theory

A theoretical limit price is useless if the book is 400 shares deep. The engine therefore computes, for your actual capital:

Field	Computation
best_ask, depth_at_ask	From live order-book endpoint, watchlist names only, 15-minute cadence
entry_ladder	Tranche 1: 40% at MIN(buy_below, best_ask) · Tranche 2: 35% at buy_below × 0.98 · Tranche 3: 25% at buy_below × 0.95. Tranches 2 and 3 conditional on the thesis remaining intact.
days_to_build	target_position_value ÷ (0.15 × ADV_20d)
est_slippage	Amihud-scaled impact for your specific order size
all_in_entry_cost	limit price × 1.0112 + est_slippage
breakeven_price	Entry grossed up for both-way costs. Displayed always. Seeing "you need +2.4% just to break even" changes behaviour more reliably than any warning label.

Limit orders only. Never market orders on this exchange — the book is too thin and you will pay 2–3% in a single fill. If `days_to_build > 10`, the interface flags it plainly: *good company, wrong size for you*. The liquidity cap will bind more often than expected, and that is the system working correctly.

27.1 Position sizing — take the minimum of four caps

```
size = MIN(
  risk_parity      = (target_portfolio_vol × risk_budget) ÷ residual_vol,
  liquidity_cap    = (0.15 × ADV_20d × acceptable_exit_days) ÷ portfolio_value,
  concentration_cap= tier_max (10% / 6% / 3%),
  sector_cap       = 30% sector – existing sector exposure
)
```

28 • Exit framework — five independent triggers

Any one fires. The system distinguishes carefully between five different reasons to sell, because conflating them is how good positions get sold early and bad ones get held forever.

Trigger	Rule	What it actually means
Valuation	Price enters the Exit zone (FV × 1.15) on the blended anchor	The business did not change; the price did. Take the money.
Thesis break	Any new integrity red flag, or fundamental score falls below 45, or a falsification condition you wrote at purchase is met	Exit regardless of P&L. The reason you bought has stopped being true. This is the most important trigger and the hardest to obey.
Momentum break	Timing verdict flips to exit, or price closes below the 100-day moving average on above-average volume	The market may know something before the filings do. Reduce, do not necessarily eliminate.
Macro regime flip	Regime shifts to Risk-Off with probability >0.7	Reduce all positions to 50% of target. Not a view on any company.
Time stop	24 months with no thesis progression (extended from 12, given the 12–36 month horizon)	Capital has an opportunity cost. Being right eventually is a form of being wrong.

NO FIXED PERCENTAGE STOP-LOSS

On an illiquid market where a stock can move 5% on a few hundred thousand rupees of volume, a fixed stop is a guaranteed donation to whoever is watching the book. Use thesis-based and structural exits. Stops are appropriate only for explicitly designated trading positions, never for long-horizon holdings.

Four concepts, never conflated in the interface: *Fair value* is the estimate of intrinsic worth. *Price target* is where the valuation is expected to be in twelve months given forecast earnings. *Take-profit zone* is where valuation becomes stretched relative to that. *Thesis break* is a reason to exit even if none of the price levels has been reached.

Part G • Macro & Sector Engine

On the CSE this layer is worth more than the other three combined, because Sri Lankan equities move as a bloc with the rate cycle far more than they move on idiosyncratic news. For a 12–36 month horizon it is decisive.

29 • Variable set

Block	Series
Monetary	Overnight Policy Rate, AWPLR, 91d/182d/364d T-bill yields, 2y/5y/10y bond yields, M2b growth, private-sector credit growth, sector-wise credit
Prices	CCPI headline and core y/y, NCPI, GDP deflator, wage index
External	LKR/USD, LKR/EUR, LKR/INR, NEER, REER, gross official reserves, months of import cover, trade balance, tourist arrivals and earnings, worker remittances, Brent
Fiscal / sovereign	Primary balance, revenue-to-GDP, debt-to-GDP, sovereign spread, rating actions, IMF review milestones and structural benchmarks, budget allocations by ministry
Real economy	GDP by sector, PMI, cement and steel consumption, vehicle registrations, electricity generation, port throughput, apparel and tea export volumes and unit prices
Market internals	ASPI turnover, foreign net flow, advance–decline, market-wide earnings yield, earnings yield minus 364-day T-bill spread
Global	US Fed policy rate and path, US 10-year, DXY, EM risk proxies, tea/rubber/palm oil prices, freight rates

THE HERO VARIABLE

CSE equity is priced as a substitute for Treasury bills. When the 364-day yield falls, domestic money rotates into equities and the market re-rates *regardless of earnings*. The **equity earnings yield minus 364-day T-bill yield** spread is therefore the single most powerful macro variable in the system. The recent sequence proves it: the OPR was raised 100bp on 26 May 2026 and the ASPI fell roughly 800 points over the following days with no change whatsoever in corporate fundamentals. Build the macro layer around this spread, and put it on the home screen as the hero chart.

30 • Method chain

Execute in this order. Each step's output is stored and displayed.

Step	Method	Purpose and Sri Lanka-specific note
1	Stationarity and break testing — ADF, Phillips–Perron, KPSS, then Zivot–Andrews with an endogenous break	The 2022 sovereign default is a structural break in nearly every series. Ignoring it makes you conclude "no cointegration" when a relationship exists in both sub-periods.
2	Estimator selection by integration profile	All I(1) with Johansen cointegration → VECM . Mixed I(0)/I(1), none I(2), short sample → ARDL bounds test (this will be the default). No cointegration → VAR in first differences.
3	Dynamics — impulse response functions (12–24 months), forecast error variance decomposition, Toda–Yamamoto Granger causality	Cholesky ordering justified by economic exogeneity: global → policy → domestic real → financial → market. Toda–Yamamoto is valid regardless of integration order; use it in place of standard Granger tests.
4	Regime classification — two- or three-state Markov switching on market returns and volatility, augmented with a macro composite z-score	Report the probability , not just the label. A 55/45 split is a fundamentally different instruction from 90/10, and the interface must show that.
5	Event study — cumulative abnormal returns, estimation window –250 to –30, event window –5 to +20	Wider than developed-market convention because information diffuses slowly here and the post-event drift is where the tradable edge sits. Event classes: CBSL policy decisions, CCPI releases, IMF review outcomes, budget presentations, elections, rating actions, tax changes, FX moves >3% in a month.
6	Sector sensitivity matrix — regress each sector's excess return on macro shocks	The practical output that drives sector tilt (\$33).

Why ARDL is the right default. Sri Lankan macro series are a genuine mix of I(0) and I(1); the usable post-liberalisation sample is short; and ARDL handles small samples better than Johansen. It also delivers the two things you actually want from one estimation: a long-run cointegrating relationship (where *should* the market be, given the macro state?) and an error-correction term (how fast does it get there?). An ECT of –0.28 on monthly data means about 28% of the gap closes per month — a half-life of roughly 2.1 months, which is directly actionable as a holding-period expectation.

31 • Regime classification and its consequences

Regime	Signature	Gross exposure	MoS add	Style tilt
Risk-On (reflation)	Falling yields, stable LKR, positive credit growth, wide earnings-yield spread, reserves accumulating	85–100%	+0%	Momentum, small cap, higher beta, cyclicals
Transition	Mixed signals, rising uncertainty, conflicting internals	50–70%	+5%	Quality, large cap
Risk-Off	Rising yields, LKR depreciation, reserve drain, spread compression, inflation above target	20–40%	+12%	Quality, low leverage, defensive sectors, USD earners, cash

The regime output is not advisory. It mechanically raises the discount rate (\$17.2), widens every margin of safety (\$25), caps gross exposure, and reweights the composite score (\$39). One estimation flows into every price in the system.

32 • Current calibration — worked example, August 2026

Illustrating how the engine should read the present moment. These are live-read values, shown here to demonstrate the logic.

Signal	Reading	Regime contribution
Policy rate direction	Tightened 100bp in May, held in July	Risk-Off
T-bill yields	Risen across the curve to ~10.1–10.3%	Risk-Off — the equity hurdle just rose
Inflation	6.8% y/y, above the 5% target, energy-driven	Risk-Off
Reserves	US\$6.45bn, drawn down from US\$6.8bn	Risk-Off
Currency	Depreciation pressure, stabilised recently after policy action	Transition
IMF programme	Combined 5th and 6th reviews completed; final programme year	Risk-On — the reform anchor holds
Fiscal	Temporary relief package plus cyclone reconstruction spending; IMF assessed the easing as appropriate	Risk-On for construction and materials; neutral overall
Growth	2026 forecast cut to ~3.0% from ~5% in 2025	Risk-Off
Market internals	ASPI -4.5% from its early-June peak; persistent foreign outflow; turnover concentrated in crossings	Risk-Off
Oil	Elevated on Middle East conflict, with second-round effects still feeding through	Risk-Off, sector-differentiated

COMPOSITE READ

Transition, probability-weighted toward Risk-Off. Recommended gross exposure in the 50–65% band, quality and low-leverage tilt, preference for USD earners and companies with pricing power, avoidance of unhedged foreign-currency borrowers and import-cost-exposed manufacturers. Margins of safety carry the +5% to +12% regime add. Note the genuine two-sidedness: the reform anchor and reconstruction spending are real positives, which is precisely why the label is Transition rather than Risk-Off, and why the probability matters more than the label.

33 • Sector sensitivity matrix

Estimated live from the regressions in §30 step 6. The structure below is illustrative of the *shape* of the output — the platform must populate it from its own estimation and never hard-code it, because these relationships shift.

Sector	Rate cut	LKR depreciation	Oil spike	Tourism rebound	Fiscal expansion
Banks	+ re-rating / – NIM	–	~	+	+ (credit growth)
Non-bank finance	++	–	–	+	+
Diversified holdings	++	mixed	–	+	+
Hotels & travel	+	++ (USD earners)	– (airfares)	++	~
Manufacturing (import input)	+	--	--	~	+
Exporters (apparel, tea, rubber)	+	++	–	~	~
Construction & materials	++	–	–	~	++
Telecom	+	– (USD capex)	~	+	~
Power & energy	+	–	– unless pass-through	~	+
Plantations	+	++	–	~	~
Consumer staples	+	–	–	+	+
Property	++	–	–	+	+

The engine translates a macro reading into a company-level implication and states it in one sentence on each stock page — for example: "Oil +18% over three months; this company imports 62% of input cost with a historical pass-through lag of two quarters; modelled EBIT margin impact –180bp for two quarters, reflected in the bear case." That is the whole point of the macro layer: not a dashboard of national statistics, but a specific, quantified line in a specific company's forecast.

34 • National project and outlook register

NEW A structured register of confirmed Sri Lankan projects and policy programmes, each mapped to affected tickers, because for a 12–36 month horizon these are the concrete catalysts.

Field	Content
Project, sponsor, sector	Name, developer, financing source, whether state, FDI or PPP
Capex and timeline	Value in LKR and USD, phase dates, current status
Status ladder	Announced → MoU → financing closed → under construction → operational. Only "financing closed" and beyond may influence a base case. Earlier stages influence only the bull case.
Affected tickers	Explicit mapping with the transmission channel — contractor, materials supplier, financier, landlord, beneficiary of demand
Quantified impact	Estimated revenue or margin effect, with the assumption stated and provenance-tagged E or F
Source and confirmation	URL, date, human confirmation required before it can affect any valuation

Current examples of register content: cyclone reconstruction allocation and its flow to construction and materials; the IMF programme's structural benchmarks and their sector consequences; electricity and fuel cost-recovery pricing restoration; tariff and trade liberalisation steps; port and logistics expansion; tourism capacity additions. Each becomes a dated, sourced row that a valuation can legitimately reference — and that a post-mortem can later check against what actually happened.

Part H • Quantitative Overlays

These layers do not select companies. They certify that an apparent opportunity is genuinely company-specific rather than a factor exposure you already own, and they help with timing at the margin. On a long horizon they matter less than Parts E and G, but they prevent two expensive errors: double-counting a bet, and buying a falling knife.

35 • CSE-native factor library

No Fama–French library exists for Sri Lanka. Building one is a durable asset — treat it as a product, not a preprocessing step.

35.1 Construction

- **Risk-free:** 91-day T-bill weekly primary market yield, converted to the return frequency.
- **MKT-RF:** free-float value-weighted total return of the investable universe, minus R_f . *Do not use ASPI* — it is full market-cap weighted and dominated by non-tradable controlling stakes, so it does not represent an achievable portfolio.
- **SMB / HML:** 2×3 Fama–French sort. Formation 30 September. Size split at universe median free-float market cap; book-to-market split at 30th/70th percentiles using the most recent audited annual report.
- **HML_hard:** a second book-to-market computed on *hard book* — revaluation reserves on land and buildings stripped for hotel, plantation and property companies. Property-revaluation-inflated book value is the number-one source of fake "value" on this exchange. Carry both; **use HML_hard as primary**.
- **MOM:** 2×3 sort on size × prior 12-month return skipping the most recent month (t–12 to t–2), rebalanced monthly.
- **LIQ:** long the least-liquid tercile, short the most-liquid tercile by Amihud, within size groups. **Not optional here.** On an illiquid frontier market the liquidity premium is often larger than HML, and omitting it means it masquerades as alpha.

35.2 The non-synchronous trading correction

CSE stocks routinely go days without a trade, which biases standard OLS betas severely downward. Every beta must use the **Dimson (1979) aggregated-coefficient method**:

$$R_{i,t} - R_{f,t} = \alpha + \sum_{k=-1..+1} \beta_k \cdot \text{MKT}_{(t+k)} + [\text{same for SMB, HML, MOM, LIQ}] + \varepsilon$$
$$\beta_{\text{true}} = \beta_{-1} + \beta_0 + \beta_{+1}$$

Skipping this is the most common technical error in frontier-market factor work, and it makes illiquid small caps look like low-beta free lunches. Newey–West standard errors, lag 4.

35.3 Estimation windows

Primary: weekly returns, 156-week (3-year) rolling window. Secondary: monthly returns, 60-month window, reported alongside for stability comparison. **Any regression spanning March 2022 – December 2023 carries a structural-break flag in the interface.** Never silently pool pre- and post-default data.

36 • Carhart certification — the referee

Job: decompose historical return into paid-for factor exposure versus genuine unexplained return.

Output	Use
α annualised, with t-statistic	Require $t > 1.5$ to count as evidence rather than a positive point estimate
β on MKT, SMB, HML, MOM, LIQ	Style fingerprint; feeds portfolio-level exposure control
R^2	Below 0.15 the model explains almost nothing — treat α as noise and downweight this layer for that name
Residual (idiosyncratic) volatility	Primary input to position sizing
Rolling 36-month α path	Decaying α is a warning; spiky α is usually a data error

THE INTERPRETATION RULE MOST PEOPLE GET WRONG

A stock with high raw returns and $\beta_{\text{MOM}} = 0.9$ is not a good pick — it is a momentum bet you are probably already making elsewhere. Specifically: if the timing engine flags a stock on momentum *and* Carhart shows high β_{MOM} with near-zero α , the signal is **redundant, not confirmed** — downgrade it. If the fundamental engine flags a stock on value and Carhart shows high β_{HML} with near-zero α , you are buying the value factor rather than a special company — acceptable, but size it as a factor bet. **The prize case** is a stock flagged by fundamentals or timing where Carhart shows positive α with modest loadings. That is genuine idiosyncratic mispricing, and those are your highest-conviction names.

Portfolio-level use. Run the same regression on your live portfolio's return stream. If aggregate $\beta_{\text{HML}} > 1.2$ or $\beta_{\text{MOM}} > 0.8$, you have accidentally become a single-factor fund — trigger a diversification alert.

37 • Momentum and contrarian timing

Fundamentals determine *what*; timing determines *when*. Central design principle: **horizon separation**. Momentum and reversal are not competing theories; they operate at different horizons, and the engine runs both without letting them fight.

Horizon	Effect	Signal
1 month	Reversal (overreaction)	Prior 1-month return, reversed
3–12 months	Momentum (slow diffusion)	t–12 to t–2 and t–6 to t–1 cumulative return
36–60 months	Reversal (long-run over-extrapolation)	Prior 3–5 year return, reversed

Signal	Definition	Weight
52-week high proximity	Price ÷ 52-week high (George–Hwang)	20%
Residual momentum	t–12 to t–2 momentum computed on Carhart residuals, not raw returns	20%
MOM_12_2	Cumulative return t–12 to t–2	20%
MOM_6_1	Cumulative return t–6 to t–1	15%
REV_1M	–1 × prior month return	15%
Volume confirmation	20-day turnover ÷ 120-day turnover, z-scored	10%

The two highest-weighted signals are deliberate. **52-week high proximity** typically outperforms raw momentum in frontier markets — anchoring is a strong behavioural force in retail-dominated books, and the metric is far more robust to stale prices than cumulative return. **Residual momentum** strips out the factor component, leaving stock-specific drift, which is exactly what Carhart is trying to certify as alpha.

37.1 Contrarian branch — all five conditions must hold

1. REV_1M in the bottom decile (a sharp recent selloff)
2. Fundamental score ≥ 70 (the business is intact)
3. No integrity red flag
4. No negative earnings revision or adverse disclosure in the last 60 days
5. The selloff is not attributable to a sector-wide macro shock still in force per the macro engine

This is the baby-thrown-out-with-the-bathwater trade, and on the CSE it has a high hit rate precisely because retail selling is indiscriminate. Conditions 3 and 4 are what stop it becoming a falling-knife machine.

37.2 Momentum crash guard

Momentum strategies suffer catastrophic drawdowns during sharp rebounds off market bottoms in high-volatility states. Hard rule: **when the macro engine reports Risk-Off with a rising transition probability to Risk-On, momentum weight is cut to 25% and contrarian weight is raised**. The timing engine subscribes to the macro engine; it is not independent of it.

37.3 Technical indicators

Moving averages (50/100/200), RSI, MACD, Bollinger bands, ATR, support and resistance from volume profile, relative strength versus ASPI and versus sector — computed on daily, weekly and monthly timeframes. **Never used in isolation, and never as a buy reason**. Their role is confirmation, entry refinement and exit warning. The convergence case — fundamentally undervalued, improving earnings, positive price momentum, rising volume — receives a higher confidence score, and that is the only way a technical signal changes an outcome.

Part I • Scoring & Ranking

38 • Composite investment score

Configurable weights, shown as a stacked breakdown that expands to inputs at every level.

Component	Weight	Contents
Valuation	25	Discount to blended fair value, adjusted for dispersion; justified-multiple residual; own-history percentile; reverse-DCF plausibility
Business quality	25	ROIC-WACC spread, gross profitability, margin stability, cash conversion, F-Score, competitive position
Growth	15	Fundamental growth (reinvestment $\times$ ROIC), revenue/EPS/FCF trajectory, growth quality and durability
Financial strength	10	Net debt/EBITDA, coverage, Altman Z", refinancing cliff, FX exposure
Macro & sector fit	10	Sector sensitivity to the current regime, project-register exposure, sector momentum
Timing & momentum	10	Timing battery score and branch label
Risk	5	Residual volatility, liquidity band, concentration of revenue or counterparty, governance
Integrity — not scored. A hard veto. Any red flag caps the name out of the buy list regardless of every other number.		

Rendered as: SAMP.N0000 — 81/100 with each component's contribution and direction arrow, every one expandable to the underlying ratios, and every ratio expandable to its source document and date. **The user must be able to reach the audited PDF page from the score in three clicks.**

39 • Fusion — sequential gating, not averaging

The layers answer different questions and are not comparable quantities. Averaging them destroys information. Fusion is sequential, and the order matters.

```
STAGE 1 — MACRO sets the frame
→ gross exposure, style weights, sector tilts, MoS regime add
→ If Risk-Off with P > 0.7: cap new entries, raise quality weight,
  no new small-cap positions

STAGE 2 — FUNDAMENTALS + VALUATION build the longlist
→ Keep composite  $\geq 65$  (sector-relative) AND price in Accumulate zone
→ Hard veto on any integrity red flag, regardless of other scores
→ Apply macro sector tilts to the ranking
→ Result: 20–35 names

STAGE 3 — TIMING orders the entry
→ 'wait' names go to a watchlist with a trigger price, not the buy list
→ Result: 8–15 actionable names

STAGE 4 — CARHART certifies and sizes the risk
→ Reject names where the whole signal is explained by factor loading
  ( $\alpha$  t-stat < 0.5 AND dominant  $\beta$  aligns with the flagging layer)
→ Supply residual volatility for position sizing
→ Result: 5–10 ranked, conviction-tiered candidates
```

Regime-conditional weight adjustment: in Risk-On, timing weight $\times 1.3$ and valuation weight $\times 0.85$; in Risk-Off, valuation weight $\times 1.3$, quality sub-pillar $\times 1.5$, timing weight $\times 0.6$.

39.1 Conviction tiers and the profitability floor

Tier	Requirement	Target position
Tier 1	Composite ≥ 80 , α t-stat > 1.5, all layers aligned, expected 12-month return $\geq 20\%$	8–10% of portfolio
Tier 2	Composite 70–80, three of four layers aligned, expected 12-month return $\geq 14\%$	4–6%
Tier 3	Composite 60–70, expected 12-month return $\geq 10\%$	2–3%
Reject	Expected 12-month return < 10%	—

THE FLOOR IS NOT ARBITRARY

With 3–5% realistic all-in round-trip friction on a mid-cap and a 364-day T-bill paying above 10% risk-free, an equity investment expected to return less than 10% over twelve months is a losing trade *even when the model is right*. You would earn more in Treasury bills with none of the risk. This single rule will do more for net returns than any refinement to any model in this document.

40 • Opportunity ranking

Continuously updated, and ranked by **risk-adjusted expected return net of the cost of building the position** — not by raw upside, which systematically flatters illiquid micro-caps you could never actually own.

```
rank_score = (expected_return_12m - all_in_round_trip_cost)
              ÷ residual_volatility
              × conviction_multiplier
              × liquidity_feasibility_factor
```

Rank	Stock	Score	Fair value	Buy below	Price	Upside to FV	Days to build	Tier
1	Company A	87	450	324	310	+45%	4	Tier 1
2	Company B	82	280	207	215	+30%	7	Tier 2
3	Company C	78	190	140	160	+19%	2	Tier 2

Structure only — populated live. Every row expands to the full evidence panel. Separate ranked lists are maintained for Core and Watch coverage tiers, and a diff view shows what changed since yesterday and why.

Part J • Portfolio & Monitoring

41 • Portfolio engine

Supports both a live portfolio and unlimited virtual portfolios, so a strategy can be paper-run alongside the real one without contaminating the decision record.

Position level

- Holdings, average cost, current price
- Unrealised and realised P&L, gross and net of all fees
- Dividend income received and accrued
- Total return including dividends
- Holding period, and time remaining to the time stop
- Current price position on the ladder
- Distance to each of the five exit triggers
- Maximum adverse and favourable excursion since entry

Portfolio level

- Allocation by position, sector, archetype, liquidity band
- Portfolio-level Carhart regression: your *true* factor exposure versus target
- Portfolio beta, residual volatility, contribution to risk by position
- Concentration and crowding: pairwise correlation of composite scores across holdings
- Gross exposure versus the regime recommendation
- Days-to-exit for the whole book at 15% of ADV
- Performance versus ASPI, S&P SL20 and an equal-weight universe benchmark

42 • Thesis-drift monitor

CENTRAL TO THE WHOLE DESIGN For every holding the system continuously answers one question: **has the reason you bought this stopped being true?** It can do this only because the decision record froze the full model state at purchase (\$45).

Drift check	Comparison against the frozen decision record
Valuation drift	Fair value now versus fair value at decision, decomposed into: earnings revision, discount-rate change, multiple change, macro regime change
Fundamental drift	Each pillar score now versus at decision; any pillar down more than 15 points is flagged
Integrity drift	Any new red flag — automatic exit trigger
Falsification check	The condition you wrote in "what would prove me wrong?" evaluated against current data. This is the highest-value automated check in the platform.
Macro drift	Regime at decision versus now; sector fit then versus now
Relative drift	Has a better risk-adjusted opportunity appeared? Ranked alternatives shown with the switching cost calculated, because a 3% better opportunity is not better after 5% of round-trip friction.

Presented as a one-line verdict per holding — *thesis intact* · *thesis weakening* · *thesis broken* — with the drivers expandable. Not as an alarm.

43 • Alerts

Alert	Trigger	Delivery
Buy-below crossed	Watchlist name enters the Accumulate zone	Push — the only genuinely time-sensitive alert in the system
Exit trigger fired	Any of the five triggers on a held name	Push
Announcement on a held or watchlist name	New CSE disclosure, classified and triaged	Push, with the AI classification and horizon tag
Thesis weakening	Pillar or falsification drift	Daily digest
Regime change	Markov transition probability moves >0.25 in a month	Push + dashboard banner
Data quality	Reconciliation mismatch, stale source, ticker quarantine	Daily digest, and a persistent badge on affected names
Factor drift	Portfolio $ \Delta\beta > 0.4$ versus target	Weekly

ALERT DISCIPLINE

Everything that is not a crossed price or a fired trigger goes into a digest. A 12–36 month strategy that generates daily push notifications will push you into overtrading, and overtrading in a 2.24%-round-trip market is precisely how the cost structure kills you. **The alerting design is a risk control, not a feature.**

Part K • AI Layer & Self-Improvement

44 • The research writer

For every company, on every material update, the language model reads the deterministic outputs and writes a structured research note. It receives numbers; it produces prose. It never produces a number that feeds back into a calculation.

Section	Content and constraint
Investment thesis	Three to five sentences. Must reference at least three specific figures with their periods.
Key strengths	Ranked, each tied to a named metric and its trend
Key weaknesses	Same discipline. If none are found, it must say so explicitly rather than omit the section.
Catalysts	Dated where possible: earnings dates, project milestones, policy decisions, regulatory resets
Risks	Company-specific, sector-specific and macro, each with the transmission channel named
Valuation	The anchors, their spread, and a plain-language reading of <i>why</i> they disagree
Buy zone / exit zone	Restated from the price engine with the margin-of-safety components spelled out
Thesis breakers	Specific, checkable conditions — "net interest margin below 3.4% for two consecutive quarters", not "if fundamentals deteriorate"
What the models disagree about	NEW Explicitly names every conflict between layers and states the historical base rate for that pattern where sample allows

CONSTRAINTS ON THE WRITER

Every quantitative claim must be traceable to a stored field, and the note is rendered with inline provenance chips. The model is instructed to write "data unavailable" rather than infer. Notes are versioned and diffable, so you can see what changed in the machine's reading of a company between quarters — which is often more informative than the note itself. **No verdict word appears anywhere.** The note describes; it does not conclude.

45 • The decision record — why this ships early

Freezing the full model state at decision time is what makes learning possible. Without it you can never answer the only question that matters for improvement: *was the model wrong, or was I?*

```
decisions (
  decision_id, ticker, timestamp, config_hash,
  action,          -- buy / watchlist / pass / partial / sell / trim
  size_pct, limit_price, conviction_1_5,
  reasoning_text,
  falsification_text, -- "what would prove me wrong?"
  -- full model state frozen at decision time:
  fundamental_score, pillar_scores_json, integrity_flags_json,
  fv_by_method_json, fv_blended, dispersion, mos_components_json,
  buy_below, fair_value, trim_above,
  timing_score, timing_branch, timing_signals_json,
  macro_regime, macro_prob, sector_fit,
  alpha, alpha_tstat, betas_json, residual_vol,
  market_price_at_decision, data_completeness_pct,
  agreement_score,      -- how aligned the layers were
  override_flag         -- did the human go against the composite?
)

outcomes (
  decision_id, exit_date, exit_price, exit_trigger,
  gross_return, net_return, holding_days,
  max_adverse_excursion, max_favourable_excursion,
  attribution_json      -- model / execution / macro / luck
)
```

Two fields matter more than they look. **Conviction 1-5** lets you later test whether your own confidence is calibrated — most people's is not, and finding out is worth a great deal. **"What would prove me wrong?"**, written *before* the trade, is what makes post-mortems honest instead of retrospective storytelling, and it is what the drift monitor evaluates automatically thereafter.

SHIP THIS IN PHASE 3, BEFORE MOST MODELS EXIST

Every decision you make without a recorded rationale is a data point you can never recover. The agent's future usefulness is a direct function of how early you start logging. Start recording the day you can see a price and a score, even if the score is only the fundamental one.

46 • The supervisory agent — three tiers, earned

This is the component you are most interested in and the one most likely to lose money if built carelessly. A language model that retunes a system based on recent results will, left unconstrained, chase noise and change parameters at exactly the wrong moments — after losses, which is when reversion is most likely.

Tier A — Narrator (read-only • deploy from day one • zero risk)

Function	Output
Pre-decision brief	Plain-language synthesis of all layers for a candidate, explicitly naming where they conflict and what each conflict has historically implied
Regime commentary	Translates the ARDL and impulse-response output into readable prose: what changed, why, what it implies for exposure
Disclosure triage	Reads new CSE announcements, classifies them (earnings / corporate action / related-party / governance / capital raise), tags sentiment and horizon, flags anything touching a held or watchlist name within minutes
Anomaly surfacing	"This name moved +14% on 4× normal volume with no announcement" — surfaces, does not interpret
Post-mortem draft	On every closed position: what the models said, what you said, what happened, and a first-pass attribution
Filing extraction	Pulls structured fields from PDF financial statements into the human confirmation queue, with page anchors

Tier A is where most of the agent's real value lives. Language models are genuinely excellent at synthesis, translation and pattern-narration across messy text, and CSE disclosure is exactly that. Lean on this hard.

Tier B — Diagnostician (read + propose · from month 3 · proposals only)

Reviews the accumulated decisions and outcomes tables and answers questions you cannot easily see yourself:

- Which layer has the highest rank information coefficient on your *actual closed trades*, versus in backtest?
- **Do your overrides add or destroy value?** Compare returns where you went against the composite versus with it. This is the single most valuable analysis the agent can run.
- Is your conviction calibrated? Do 5s outperform 3s?
- Where do losses cluster — which sector, score band, regime, liquidity bucket, data-completeness band?
- Are entries systematically early or late? Consistently large maximum adverse excursion before profit means the entry ladder is too aggressive.
- Is any layer decaying? Rolling information coefficient versus backtest.

Output format is a **proposal ticket**, never a change:

PROPOSAL #14

Observation : 12 of your 15 losing trades since January had an integrity score between 50 and 65. Winners averaged 79.

Sample : 38 closed positions, 14 months.

Hypothesis : The integrity veto threshold is too permissive.

Proposed : Raise hard-veto floor from 45 to 60.

Backtest : [Tier C runs this]

Risk : Reduces universe by ~11 names; may cut valid opportunities.

Status : AWAITING HUMAN REVIEW

Tier C — Optimiser (propose + auto-validate · month 12+ · still cannot deploy)

Can execute a proposal through the full validation harness — purged walk-forward, cost model, Deflated Sharpe, baseline comparison, regime-by-regime breakdown, parameter sensitivity sweep — and present the results. **It cannot merge. You merge.**

47 · Guardrails — non-negotiable

#	Guardrail	Why
1	Minimum sample: 30 closed positions <i>and</i> 12 months before any proposal is even generated	Below this, every "pattern" is noise. This rule alone prevents most of the potential damage.
2	Parameter freeze during any drawdown >15%	Retuning after losses is the classic wealth-destroying reflex. The agent is explicitly forbidden from proposing changes while you are hurting.
3	Change budget: one parameter change per quarter, maximum	Rate-limiting forces prioritisation and keeps results attributable. Ten simultaneous changes teach you nothing.
4	Every proposal clears the same bar as an original model	Purged walk-forward, Deflated Sharpe > 0, must beat the simple baseline. No exceptions for agent-authored changes.
5	The cost model, liquidity caps, position limits, expected-return floor and integrity veto are constitutional — the agent may never propose changes to them	These are survival constraints, not optimisable parameters. An optimiser will always want to relax them, because relaxing them improves backtest returns.
6	Full config versioning — every config git-tracked with a hash; every signal and decision stamped with the hash that produced it; one-click rollback	Without this you cannot tell which version generated which result.
7	The agent is scored too	Track realised performance of its accepted proposals. If they underperform the unchanged system over a year, demote it to Tier A.
8	The agent never has write access to the database or to order flow	It writes tickets. Humans write parameters. Nothing writes orders.

THE HONEST ASSESSMENT

Claude will be very good at Tier A and genuinely useful at Tier B. Tier C is where to move slowly, and the reason is not capability — it is sample size. With 15–25 round trips a year you accumulate statistically meaningful evidence about your own system at a rate of roughly one solid conclusion every eighteen months. No amount of model intelligence compresses that. **The agent's most valuable contribution over the first two years will be making you honest about your own decisions, not tuning coefficients.**

Part L • Validation

48 • Backtest protocol

You will generate false alpha unless every one of these is enforced. This section is the difference between a system that works and a system that looks like it works.

#	Requirement	Detail
1	Point-in-time only	Every query filters on <code>first_available_date</code> . Restated financials never enter.
2	Survivorship-inclusive	Delisted and suspended names carry terminal returns.
3	Purged walk-forward cross-validation	Train on window W , purge the following period equal to the forward-return horizon to prevent label leakage, then test. Never random k -fold on a time series.
4	Realistic cost model, always applied	1.12% each way plus a slippage term scaled to Amihud illiquidity and order size relative to ADV. Report gross and net side by side, and treat net as the only real number.
5	Deflated Sharpe Ratio	You are testing dozens of variants. Report the DSR corrected for the number of trials. If $DSR \leq 0$, you have found nothing.
6	Regime coverage	Must span at least three distinct macro regimes, with performance reported per regime. A strategy that only works in reflation is a leveraged bet on rate cuts.
7	Capacity curve	Re-run at LKR 5m / 25m / 100m / 500m. Plot alpha decay against capital. This tells you the size at which your edge dies.
8	Sensitivity sweep	Vary every key parameter $\pm 30\%$. If performance collapses you have curve-fitted. A robust strategy is a plateau, not a peak.
9	Baseline comparisons — mandatory	Against ASPI buy-and-hold, S&P SL20 buy-and-hold, equal-weight universe, and a simple fundamental z -score screen alone. If the full stack does not beat the simple screen out of sample, use the simple screen. Complexity must earn its place.
10	Paper-trade two quarters before capital	Log every signal with a timestamp. Expect 50–70% of backtest performance to survive contact with reality. If you get more, suspect a bug.

Measured: CAGR, total return, win rate, average win $\div$ average loss, maximum drawdown and its duration, Sharpe, Sortino, volatility, profit factor, turnover, and performance relative to each benchmark — all reported gross and net.

49 • Target metrics and kill thresholds

Metric	Target	Kill threshold
Hit rate, 12-month horizon	$\geq 58\%$	$< 50\%$ over 20 closed positions
Average win $\div$ average loss	$\geq 1.8\times$	$< 1.2\times$
Net annual return versus ASPI	+10–15pp	Underperform two consecutive years
Annual turnover	$\leq 100\%$	$> 200\%$ (costs eat you)
Maximum drawdown	$\leq 25\%$	$> 35\%$
Live information ratio $\div$ backtest IR	≥ 0.5	< 0.3 over 12 months

50 • Monitoring and kill switches

Run nightly; alert on breach.

Monitor	Metric	Alert threshold
Data integrity	Corporate-action reconciliation mismatch	$> 0.5\%$ on any ticker $\rightarrow$ quarantine
Data freshness	Hours since last successful ingest per source	$> 36h$
Signal decay	Rolling 12-month rank IC of each layer	Below 50% of backtest IC
Regime stability	Markov transition probability	Change > 0.25 in one month
Factor drift	Portfolio β versus target	Any $ \Delta\beta > 0.4$
Live versus backtest	Rolling 12-month IR ratio	< 0.5 review $\cdot < 0.3$ halt new entries
Crowding	Composite score correlation across held names	Pairwise mean > 0.7
Capacity	Position size versus current ADV	Days-to-exit > 10

51 • Layer map and stack

L9	DECISION SURFACE (React) Home · Screener · Stock file · Valuation workbench · Portfolio · Macro · Journal · Backtest lab · Data health
L8	AI LAYER – research writer · triage · supervisory agent
L7	MONITORING – drift, decay, kill switches, alerting
L6	EXECUTION LOGIC – price ladder, entry, exits, sizing
L5	SCORING & FUSION – sequential gating → composite
L4	ENGINES – fundamental valuation macro timing risk
L3	FACTOR LIBRARY – CSE-native MKT, SMB, HML_hard, MOM, LIQ
L2	COVERAGE & GATES – tiers, liquidity, structure, integrity
L1	DATA – point-in-time store, provenance, corporate actions

Data flows up. Constraints flow down.

Layer	Choice
Ingestion	Python + Apify actors (already operational) + PDF extraction pipeline
Storage	PostgreSQL with TimescaleDB for price series
Transform	dbt — testable, versioned, lineage-tracked
Econometrics	statsmodels (ARDL, VECM, VAR, Markov switching), arch, linearmodels
Valuation	Pure-Python deterministic modules, one per method, each unit-tested against a hand-worked reference case and against Damodaran's published spreadsheet outputs
Machine learning	scikit-learn Elastic Net, lightgbm, purged cross-validation. Heavily regularised — with ~130 investable names and ~12 usable years you are in small-data territory and unregularised ML will overfit spectacularly.
Orchestration	Prefect, or cron for v1 — do not over-engineer
API	FastAPI
Frontend	React + Tailwind + Recharts
Cache	Redis for computed scores and session state
AI	Claude API, called by the worker for Tier A/B tasks and for the research writer

52 • Scheduler — what "always on" actually means

The CSE trades roughly 09:30–14:30 local, five days a week — about 25 hours out of 168. Your horizon is 12–36 months. What runs continuously is the scheduler, the monitor and the overnight batch, not a firehose of redundant polling. Re-running an ARDL model every five minutes is theatre; the underlying data is monthly.

Job	Schedule	Purpose
Market status heartbeat	Every 5 min, always	Cheap check that gates everything else
Price snapshot	Every 5 min, market hours only	Live prices for the gap-to-entry display
Buy-below trigger check	Every 5 min, market hours	The only intraday computation that matters: has any watchlist name crossed its limit? A comparison, not a computation.
Order book	Every 15 min, watchlist names only	Depth for the entry ladder
Announcements	Every 15 min, 08:00–18:00	Corporate actions, earnings and new listings land here — the highest-value feed in the system
EOD snapshot + adjustment	15:00 daily	Close, volume, corporate-action adjustment, reconciliation test
Nightly model batch	20:00 daily	Recompute scores, run all valuation methods, refresh fair values and the price ladder, factor returns
Research note regeneration	Nightly for changed names only	AI writer, diffed against previous version
Factor regressions	Weekly, Sunday 02:00	3-year rolling window re-estimation
Financial statement ingestion	Event-driven on filing	Extraction → human confirm queue → promotion to Reported
CBSL macro poll	Release-calendar driven — T-bill auctions Wednesdays, CCPI end-of-month, policy decisions ~8/yr	Event-driven, not continuous
Macro re-estimation	Monthly, plus immediately on a policy decision or IMF milestone	ARDL / VECM / regime refresh
Full validation re-run	Quarterly	Drift and decay check

53 • Deployment

```

Single VPS (2 vCPU / 4GB – Hetzner, DigitalOcean or similar)
├─ PostgreSQL + TimescaleDB ← point-in-time store
├─ Redis ← computed scores, session cache
├─ FastAPI ← API for the React frontend
├─ Prefect / APScheduler ← the job table above
├─ Worker process ← model batch + AI calls
└─ Caddy or nginx + TLS

```

```

Frontend : React static build (Vercel / Netlify free tier)
Alerts   : Telegram bot or email
Backups  : Nightly pg_dump to object storage, 30-day retention
Secrets  : Environment-injected, never in the repo

```

Realistic running cost: **US\$25–60/month** including VPS, storage and moderate Claude API usage. If a design decision pushes this toward hundreds a month, it is probably the wrong decision at this stage.

54 • Build sequence

Ship something usable at every phase. The most common failure mode for a project like this is building every model on dirty data and never shipping.

Phase	Deliverable	Weeks	Gate to proceed
1	Point-in-time data spine, cse.lk ingestion, corporate actions with human confirm, reconciliation tests, coverage tiers, provenance model	3–4	Reconciliation passes 30 consecutive days
2	Fundamental engine, ratio history with trend detection, sector routing, integrity veto, ranked screener UI, stock file v1	4–5	Baseline screen shows positive rank IC out of sample
3	Valuation engine — model router, discount rates, DCF, residual income, justified multiples, SOTP, triangulation. Price ladder. Margin-of-safety engine.	5–6	Fair values reproduce hand-worked reference cases exactly
4	Scheduler, always-on service, alerting, home dashboard, decision capture	2–3	You are recording decisions with frozen state
5	Macro engine — ARDL, regime classifier, sector sensitivity, project register, regime-conditional discount rates and margins of safety	3–4	Regime classifier correctly labels known historical periods
6	Factor library, Carhart certification, timing engine, full fusion, complete decision card	3–4	Full stack beats the simple screen out of sample
7	AI research writer + agent Tier A (narrator, triage, post-mortems)	1–2	Notes are accurate and provenance-clean on a 20-name audit
8	Portfolio, thesis-drift monitor, backtest lab, monitoring, kill switches	3–4	Two quarters of paper trading
9	Agent Tier B, then Tier C — gated on 30 closed positions and 12 months	—	—

TWO DELIBERATE ORDERING CHOICES

Valuation moves to Phase 3, ahead of macro and factors, because your central requirement is a fair value and a buy price for every stock, and that is the shortest path to it. **Decision capture ships in Phase 4**, before most models exist, because every decision made without a recorded rationale is a data point you can never recover.

Part N • Failure-Mode Register

#	Failure mode	Why it is likely here	Mitigation
1	Look-ahead bias via restated financials	CSE restatements are common and quietly overwritten	Immutable versioned point-in-time store; <code>first_available_date</code> on every query
2	Downward-biased betas from non-trading	Structural feature of this exchange	Dimson lead/lag aggregation, mandatory; bottom-up beta blend
3	Fake value from revalued property	Endemic in hotels, plantations, property	Hard book as primary; both figures always shown
4	Backtest alpha destroyed by costs	~2.24% round-trip fees plus spread and impact	Cost model in every backtest; expected-return floor; turnover cap
5	Overfitting on small N	~130 investable names, ~12 usable years	Regularisation, purged walk-forward, Deflated Sharpe, baseline-must-be-beaten rule
6	Structural break contamination (2022 default)	Breaks every long-window estimate	Zivot-Andrews; break flags in the UI; sub-period reporting
7	Wrong valuation model for the business	Banks, conglomerates and plantations each break the standard DCF differently	Archetype router; firm-side models hard-suppressed for financials
8	False precision from a single fair value	Every input is an estimate	Multiple anchors, dispersion band, scenarios, Monte Carlo, reverse DCF
9	Momentum crash on the rebound	Classic, and violent in frontier markets	Regime gate on momentum weight
10	Signal double-counting across layers	The layers share underlying data	Carhart redundancy check at fusion Stage 4
11	Owning what you cannot sell	Illiquidity is the moat and the trap	Liquidity cap in sizing; days-to-exit monitor; feasibility factor in ranking
12	Related-party value extraction	Family-controlled conglomerate structures	Integrity hard veto, never a scored input
13	Silent data source breakage	Scrapers break when sites change	Schema validation, freshness monitors, reconciliation, quarantine on failure, second source
14	AI hallucinating a financial figure	PDF extraction is genuinely hard	AI-assisted tier cannot enter a valuation until human-confirmed; page anchors mandatory
15	Agent chasing noise	Small sample, strong incentive to act	The eight guardrails in §47; constitutional parameters; drawdown freeze
16	Human overriding correct signals	The most common real-world failure	Journal every override; quarterly review of override P&L versus model P&L
17	The interface manufacturing urgency	Any market UI drifts this way	Alert discipline (§43); calm design system (companion document); no live-flashing tickers

Part O • Open Parameters

Decisions required from you before Phase 1. Each materially changes the build.

#	Question	What it determines
1	Capital base	Liquidity thresholds, universe size, position caps, capacity curve. LKR 5m and LKR 100m are genuinely different systems — at the small end almost everything is investable and concentration is the risk; at the large end liquidity binds constantly and half the ranking is unreachable.
2	Historical data depth	How far back can point-in-time fundamentals be reconstructed? This caps your backtest and therefore your confidence. Realistically 2015-onward may be clean and earlier patchy.
3	Broker execution	Manual order entry or a broker API? Determines whether entry ladders are executed by you or by the system.
4	Tax treatment	Capital gains treatment and your residency position both affect net returns and optimal holding period. Sits in the same equation as the cost model.
5	Second data source	Identify one now for the nightly cross-check. A single unofficial endpoint is a single point of failure.
6	Dividend policy	Are you optimising total return or income? Changes the weighting between dividend-based and cash-flow-based anchors.
7	Concentration appetite	8–15 positions is the design assumption. Fewer raises single-name risk; more dilutes the edge and raises the cost drag.
8	Sector exclusions	Any sectors, structures or specific names you will not own regardless of score — encoded as a gate rather than remembered.

Part P • Appendices

P1 • Formula quick reference

ROIC = NOPAT ÷ (total debt + equity - cash)

Economic spread = ROIC - WACC

Gross profitability = gross profit ÷ total assets

Fundamental growth = reinvestment rate × ROIC

Sustainable payout = 1 - (g ÷ ROE)

FCFF = EBIT(1-t) + D&A - capex - Δ non-cash working capital

FCFE = net income + D&A - capex - ΔWC + net borrowing

Ke = Rf_LKR + β_adj × ERP_eff + size premium + illiquidity premium

WACC = Ke × E/(D+E) + Kd(1-t) × D/(D+E) [market-value weights]

Justified P/E = payout × (1+g) ÷ (Ke - g)

Justified P/B = (ROE - g) ÷ (Ke - g)

Justified EV/EBIT = (1-t)(1 - g/ROIC) ÷ (WACC - g)

Residual income V0 = B0 + Σ (ROE_t - Ke) · B_(t-1) ÷ (1+Ke)^t

Gordon growth V0 = D1 ÷ (Ke - g)

TERP = (N · P_cum + S · P_sub) ÷ (N + S)

Amihud illiquidity= mean(|return| ÷ turnover)

Dimson beta = β_(-1) + β_0 + β_(+1)

MoS = base + dispersion + liquidity + regime + quality + completeness

Buy below = FV × (1 - MoS)

Breakeven = entry × (1 + 0.0112)^2 + slippage

P2 • Archetype mapping

Every ticker carries an `archetype` field driving model selection: *bank · non-bank finance · insurance · diversified holding · manufacturing · consumer · plantation · hotel · telecom · construction & materials · power & energy · property · healthcare · logistics · other*. Mapping from GICS and CSE sector is maintained as a version-controlled file with manual overrides — several CSE conglomerates are misclassified by standard GICS and must be corrected by hand.

P3 • Glossary of terms used in the interface

Term	Plain meaning shown on hover
Fair value	What the business appears to be worth, from several independent methods
Buy below	The highest price at which you still get your required cushion
Margin of safety	Your cushion against being wrong — computed from how much the models disagree, how illiquid the stock is, what the macro environment is doing and how complete the data is
Dispersion	How much the valuation methods disagree. Wide means "we don't understand this business well."
Hard book	Book value with property revaluation gains removed — the version that reflects money actually put in
Reverse DCF	What the current share price is assuming about the future, stated as growth and margin
Residual momentum	Price drift that is specific to this company, after removing the part explained by the market and by style
Regime	What season the market is in — determines how much you should own overall
Thesis break	The reason you bought has stopped being true. Sell regardless of price.
Days to build	How many trading sessions it would take to acquire your target position without moving the price
Data completeness	What share of the inputs this analysis needs are actually available

Sources and reference frameworks. Valuation architecture and model-choice logic follow Aswath Damodaran's published framework and datasets (NYU Stern). Practitioner context on model selection, the discount-rate problem and the limits of multiples draws on the CFA Institute Research Foundation monograph *Equity Valuation: Science, Art, or Craft?* (Fabozzi, Focardi & Jonas, 2017), held in project knowledge. Factor methodology follows Fama-French and Carhart, with Dimson (1979) correction for non-synchronous trading and Newey-West standard errors. Accounting-quality signals follow Piotroski, Beneish, Sloan, Altman and Novy-Marx. Macro-econometric method follows Pesaran-Shin-Smith ARDL bounds testing, Johansen cointegration, Toda-Yamamoto causality and Markov regime switching. Validation protocol follows Bailey & López de Prado. Sri Lankan macroeconomic figures cited in \$2.3 and \$32 are as reported by the Central Bank of Sri Lanka, the IMF and Colombo Stock Exchange market data as at early August 2026, and are read live by the platform rather than hard-coded.

Nothing in this document is investment advice. It is a technical specification for a decision-support tool. The platform is designed explicitly so that a human makes every investment decision, with full visibility of the evidence and its limitations.